PAST PAPER 2022

Explain Partial correlation:

Partial correlation is a statistical method that measures the relation ship between Two variables while accounting for the effects of other variable. It's also known as conditional correlation.

Partial correlation measures the strength and direction of a linear relationship between two continous variables.

Partial correlation is used when you want to know the relation ship between two variables. your variables of interest are continous and you have covariates.

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Explain . simple regression analysis:-

Simple regression analysis is a statistical method used to model the relationship between two continous variables. Here's a step by step explanation.

Assumption:

Before performing simple regression analysis, we need to check if the data meets the following assumptions.

Linearity: The relationship between the independent variable(X) and the dependent variable Y should be linear.

Independence: Each observation (~~Each~~) should be independent of the others

Homoscedasticity: The variance of the residuals should be constant across all level of X.

Normality: The residuals should be constant across level of X.

In what type of study we can use snowball sampling

Snowball sampling is the best used in studies where it's hard to find participants because they to a specific or hidden group This method relies on participants referring others who are part of the same group, helping researchers reach more people.

For Example:

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1 **Hard to reach populations:** Studies involving marginalized groups, such as undocumented immigrants or the homless

2- **Sensitive Topics:** Research on drug users, survivors of abuse, or stigmatized illness

3- **Specialized networks:** Research on niche professional groups, activists or rare diseases-

It works well when trust and personal connections are key to finding participants.

What are the sources of random error in your research?

Random error in research are

unpredictal variations that can affect the accuracy of your results. These errors happen by chance and are not caused by mistakes or biases. Here are some common ~~res~~ sources of random error in simple terms.

1 Measurement Tools
2 Human Factors
3 Environmental Conditions
4 Biological Validity
5 Sampling Issues

The errors are normal and can't be completely avoided, but you can minimize their impact by repeating measurements and using proper statistical methods.

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Why do some researchers explicity write instructions in the questionnaires?

~~Re se arche rs often write instructions in questionnaires using simple and easy to understand language to ensure clarity and improve the quality of the responses.~~

Researcher use simple language in questionnaires so everyone can understand the instructions easily.

1- **Include Everyone:** People with different education level or language skills can follow along.

2- **Avoid Confusion:** Clear instructons prevent misunderstanding.

3- **Get Honest Answers:** Simple Instructions make sure everyone answer is the same way.

4- **Save Time:** People can complete the questionnaires quickly.

Question #2

Answer the following Questions

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How sampling and non-sampling errors can be reduced to get valid data for research?

To make sure the data you collect for research is accurate, you need to reduce two types of errors

1- **Sampling Errors**

2- **Non-Sampling Errors**

1- **Sampling Errors:** These happen when the group you choose to study (sample) does not represent the whole population well.
To reduce this: Pick people randomly to avoid bias

- Include all important groups (e.g by age or gender) in your sample.

2- **Non-Sampling Errors:** These are mistakes that happen when collecting and handling data.

- Ask simple clear questions so people understand them easily.
- Train your teeth to collect data correctly.
- Check the data carefully to fix any mistakes

By doing these thing your research will give more reliable and useful results

What feature do you think a quality research design must have?

A quality search design should have these key features

1- **Simplicity:** Easy to use search bar that's visible and straightforward.

2- **Clear goal:** Know exactly what you want to find out.

3- **Right methods:** Use the best way to gather information like (surveys, experiments etc)

4- **Consistency:** Get the same results if the research is done again.

5- **Accuracy:** Make sure the research measures what it's supposed to.

6- **Control:** Make sure nothing unfair affects the results.

7- **Plan for analysis:** Know how you'll analyze and understand the data.

8- **Replicability:** Others should be able to repeat the research and get the same results

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Suppose you are conducting a primary research. Which research methods and research methodology you will choose? Explain with the help of an example.

If I were to conduct primary

research. I migh choose the survey research method with a quantitative research metheodology.

For Example:

Let's say I want to understand people's preferences for online shopping I could create a survey with questions about their shopping habits favorite online stores, and reasons for choosing online shopping. By analyzing the survey responses quantitatively. I could graw conclusions about the tends and patterns in online shopping preferences.

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Question: 1

Answer the short questions

How indeterminacy principle effect sampling?

The indeterminacy principle affect sampling by

1- **Limiting precision:** You can't know everything exactly so there's always some uncertainty.

2- **Trade-offs:** To get more precise results you might need to sacrifice some accuracy and vice versa.

3- **Uncertainty in results:** Even with lots of data there always some uncertainty in your results.

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Explain Stratified Sampling method of data collection?

In stratified sampling researchers divide subjects into subgroups called strata based on characteristics that they share (e.g race, gender, educational attainment) Once divided each subgroup is randomly sampled using another probability sampling method

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How precision can be improved in sampling?

Precision can be improved by

- Increasing sample size
- Using stratified and systematic sample.

- Reducing sampling errors through proper plannings.
- Ensuring randomness in sampling selection.

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Why do most of the researcher use item analysis approach the scale construction?

Item analysis me an that researcher check the question or item in a test to see how good they are. This helps them understand which questions are easy which are difficult and which one are best for the test. This process is very important in scale construction to ensure that the test gives accurate results.

Why studies based on probability sampling are preferred over studies done by using non-probability sampling?

Probability sampling ensures that every member of the population has a known and equal chance of selection, leading to more representative and generalizable results. Non probability sampling locks this rigor, makings it prone to bias

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Question: 2

Answer the following Questions.

How sampling help social scientist to conduct a study?

Sampling helps social scientist by allowing them to a study a smaller manageble group of people instead of trying to study an entire population. The smaller group called a sample is carefully chosen to represent the larger population. By studying the sample researchers can make conclusion or predictions about the whole group more quickly efficiently and cost efficiently

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How sampling and non error can be reduced to get valid data for research?

To get accurate data in research, its important to reduce both sampling errors and non sampling errors.

- Reducing Sampling Errors:

Sampling errors happen when the group you study does not perfectly represent the whole population

1- Choose the right sample size
2- Random Sampling
3- Stratified Sampling

- Reducing Non-Sampling Errors

Non Sampling errors comes from mistakes in collecting or handling data like wrong questions or data entry errors.

1 Clear Questions
2 Train Data Collection
3 Test Your Tools
4 Check Data Carefully
5 Avoid Bias

By carefully selecting your sample and double-checking your data collection process, you can reduce errors and get results that are closer to the truth.

Choose a research problem of your own choices and write a report on it